Implemented UEB1 core punctuation

Task 9.3.1.1 adds seven context-free cells to the existing lowercase alphabet. These rows emit ordinary US-layout host keystrokes through ZMK `&kp`; they do not send Unicode Braille.

Result	Dots	Cell	ZMK keycode
Comma ,	2	⠂	COMMA
Semicolon ;	2,3	⠆	SEMI
Colon :	2,5	⠒	COLON (Shift+semicolon)
Exclamation !	2,3,5	⠖	EXCL (Shift+1)
Apostrophe ‘	3	⠄	SQT
Hyphen -	3,6	⠤	MINUS
Full stop .	2,5,6	⠲	DOT

Authority: ICEB, *The Rules of Unified English Braille*, third edition 2024, section 7, printed page 75. The unchanged PDF and its independent reference license are documented in `references/README.md`.

All seven rows have any-context predicates. Comma/full stop retain the numeric-punctuation class (6.2.1); apostrophe retains the opening class (2.6.2); hyphen retains its boundary class (2.6.1). Other rows use the other class. Context transitions remain phase-04 work.

Dots 2,3,6 (⠦) are not assigned yet: question mark versus opening double quotation mark requires word context. Multi-cell symbols and the remaining literary signs are later tasks. No hardware acceptance is claimed.

Verification boundary

The seven mappings are checked through production chord capture, matching, and ordered dispatch with the real factory command table. Colon and exclamation include the implicit Shift bit in the ZMK keycode; neither requires a separate sticky modifier action. QBF and Toucan central-half firmware builds pass, but host and physical-device acceptance are separate.

Printable quick-reference presentation

The printable reference lives in `docs/braille-cheatsheet/`. Table bottom borders now match their top and sides, including column continuations. Braille columns and wrapped sequences are centered except in Capitals. Capitals includes the Grade 1 symbol, word, and passage indicator examples; its examples show UEB sequences only, while the short dot-7 rule remains. The separate Capitalised contractions section is omitted. All 182 contractions and 22 keyboard chords remain. These are presentation changes, not changes to implemented mappings.

Implemented specified two-cell UEB1 symbols

Task 9.3.1.2 adds the following 21 symbols, with successive cells in the listed order. Each complete sequence emits one ordinary US-layout ZMK `&kp` press/release; shifted characters use the keycode’s implicit Shift.

Result	Dots in order	Cells	ZMK keycode
Ampersand &	4 then 1,2,3,4,6	⠲⠿	AMPS
Asterisk *	5 then 3,5	⠐⠔	STAR

At `@`	4 then 1	⠈⠁	AT
Caret `^`	4 then 2,6	⠈⠢	CARET
Number sign `#`	4,5,6 then 1,4,5,6	⠸⠹	HASH
Dollar `$`	4 then 2,3,4	⠈⠲	DLLR
Percent `%`	4,6 then 3,5,6	⠨⠬	PRCNT
Tilde `~`	4 then 3,5	⠈⠔	TILDE
Forward slash `/`	4,5,6 then 3,4	⠸⠌	FSLH
Underscore `_`	4,6 then 3,6	⠨⠤	UNDER
Plus `+`	5 then 2,3,5	⠐⠖	PLUS
Equals `=`	5 then 2,3,5,6	⠐⠶	EQUAL
Less-than `<`	4 then 1,2,6	⠈⠣	LT
Greater-than `>`	4 then 3,4,5	⠈⠜	GT
Open parenthesis `(`	5 then 1,2,6	⠐⠣	LPAR
Close parenthesis `)`	5 then 3,4,5	⠐⠜	RPAR
Open square bracket `[`	4,6 then 1,2,6	⠨⠣	LBKT
Close square bracket `]`	4,6 then 3,4,5	⠨⠜	RBKT
Open curly brace `{`	4,5,6 then 1,2,6	⠸⠣	LBRC
Close curly brace `}`	4,5,6 then 3,4,5	⠸⠜	RBRC
Backslash `\`	4,5,6 then 1,6	⠸⠡	BSLH

Authority: ICEB 2024 sections 3 and 7. Per-sign sections and printed pages are recovered in `references/research/ueb-literary-signs.yaml` from the historical research and checked against the unchanged authoritative PDF.

The four first cells (4; 5; 4,6; 4,5,6) are silent strict prefixes, held without a timer. A non-continuation drops the prefix with a notice and is interpreted afresh: 4 then 1 emits `@`, but 4 then 1,3,5,6 drops 4 and emits `z`. A new prefix after a drop can itself remain pending.

Every symbol has any-context predicates. Open parenthesis, square bracket, and curly brace use the opening class (2.6.2); less-than and the other symbols use the other class. These are metadata only: context transitions and question/quotation policy remain phase-04 work. This supersedes the earlier note that these 21 multi-cell symbols are still future work.

All 21 symbols and their first-cell non-continuations are tested through production capture, matching, and dispatch, with real command-table validation, all-command recovery, split/Grade-2 build variants, pinned ZMK keycodes, 21 YAML sign fixtures and seven scoped examples. Both central-half firmware smoke builds pass. The separate printable cheatsheet is untouched; physical keyboard/VoiceOver acceptance remains user-owned.

Printable quick-reference mnemonics and layout

The printable reference in `docs/braille-cheatsheet/` is split into modules: `style.typ`, `data.typ`, `cells.typ`, `mnemonics.typ`, `tables.typ`, and `validation.typ`, with `braille-cheatsheet.typ` as the entry point. The second heading is now “UEB Grade 2”.

Tables whose inputs are letters or letter-derived signs gain Output, Input, and Mnemonic columns: Navigation and Editing, Alphabetic wordsigns, Lower wordsigns, Lower groupsigns (all three), Initial-letter contractions, Final-letter

groupsigns, and Shortforms. Mnemonics are derived from the actual cells: letter cells spell the letter, and the ten lowered a–j cells spell ↓ plus the letter. Non-letter signs stay vector braille; rows without any mnemonic show a dash. Strong wordsigns, Strong contractions, and Strong groupsigns are labeled Output and Input without a mnemonic column; Sticky modifiers is labeled Modifier and Input. Mapping data is unchanged: all 182 contractions, 22 keyboard chords, and every character cell are identical. Compilation asserts the mnemonic spellings round-trip to the exact input cells and that labeled tables have one column per header. The reference still fits two A4 pages with no overlapping or out-of-bounds text.

Printable quick-reference arrangement

The printable reference drops the “Characters & keyboard chords” heading. Capitals follows Lowercase in the first page column; Digits occupies the second column and Symbols holds the third column alone through explicit weak column breaks, so the keyboard chords begin page 2 together. Missing mnemonics leave their cell empty instead of a dash. Compilation now requires all 22 chords to share one page rather than page 1. The reference is three A4 pages with no overlapping or out-of-bounds text.

Printable quick-reference underline convention

The dedicated Symbols column is reverted: Digits and Symbols share the second page column, the keyboard chords flow onto page 1 again, and the reference is two A4 pages. Chords held with Braille Space no longer show a Space key and plus sign; their dot cell is underlined instead, in both the Input and Mnemonic columns. The plain Space row keeps its key box, dot-only chords stay unmarked, and missing mnemonics remain empty cells.

Printable quick-reference chord merge

Sticky modifiers merge into the Navigation and Editing table, whose 22 rows now end with the five modifiers. Each modifier name and input cell carries a trailing underscore: “Shift *“ with its underlined cell followed by “*”, marking that the chord holds its effect for the next input. The UEB Grade 2 heading is dropped; the Grade 2 sections follow the chord table directly.

Printable quick-reference space row

The Lowercase table drops its Braille Space row and is retitled “Lowercase letters”; the Space chord remains in Navigation and Editing. The character coverage assertion now covers printable ASCII excluding space. The reference stays two A4 pages with no overlapping or out-of-bounds text.

Printable quick-reference digit tables

The Digits table splits into “Digits · UEB 1” and “Digits · 8 dot”, each with Number, Input, and Mnemonic columns. The 8-dot digit cells are the lowered a–j letters, so their mnemonics use the down arrow: 0 is ↓j and 1 is ↓a. UEB 1 mnemonics keep the number sign before the letter. Modifier outputs and inputs end with an ellipsis instead of an underscore, and Space chords show the bordered Space key plus the dot cell again instead of an underline. A page break puts Alphabetic wordsigns at the top of page 2; the reference is three A4 pages with no overlapping or out-of-bounds text. A multiline expected-row expression in the compilation checks silently counted only its first line and is now parenthesized.

Printable quick-reference symbol pairing

Symbols precedes the digit tables: Lowercase and Capitals fill the first page column, Symbols fills the second, and a weak column break starts Digits · UEB 1, Digits · 8 dot, and Navigation and Editing in the third. Within Symbols, opening and closing pairs are adjacent: (with), < with >, [with], and { with }, so = follows the angle pair,
follows the bracket pair, and | follows the brace pair. A compilation assert pins each closing symbol to the row after its opening counterpart.

Printable quick-reference thinner cell rules

Every cheatsheet table’s internal row separators drop from 0.3pt to 0.2pt while the 1pt black top, side, and bottom frame rules and the 0.4pt Space key outlines stay unchanged. This is a presentation change only.

Printable quick-reference space symbol

Input and Mnemonic columns replace the word Space inside the bordered key box with the open-box symbol ␣ (U+2423). The Navigation and Editing Space chord keeps its name Space in the Output column, and Capitals examples show ␣ wherever their braille sequences contain a space. Rendering and validation are unchanged; this is a presentation change.

Printable quick-reference sticky ellipsis spacing

Sticky modifier outputs drop the space before the ellipsis: Shift…, Control…, Option…, Command…, and Globe… now have no space before the ellipsis. The compilation assert matches the new form.

Printable quick-reference capitals table

Capitals becomes a two-column table with Output and Input headers. Each row pairs the printed example with its complete UEB 1 sequence: A with ⠠⠁, ABC with ⠠⠠⠉⠃, ABc with ⠠⠠⠁⠃⠠⠉, A BIG CAT with ⠠⠁⠀⠠⠠⠃⠊⠛⠀⠠⠠⠉⠁⠞, b with ⠰⠃, cd with ⠰⠰⠉⠙, and b c d with ⠰⠰⠰⠃⠀⠉⠀⠙. The per-row scope headings (Letter prefix, Word prefix, and so on) and the UEB 1 mode labels are dropped; the dot-7 intro line stays. Sequences that exceed the Input column wrap inside their cells, and spaces in those sequences show the open-box symbol. The mode-example and capital-example helpers are removed. Capitals still renders exactly seven rows, checked by the compilation assertion.

Printable quick-reference continuous columns

The weak column break after Symbols is dropped. Lowercase, Capitals, Symbols, both digit tables, and Navigation and Editing now flow continuously through page 1′s three columns: Symbols continues at the top of column 2 as “Symbols Cont’d”, and Navigation and Editing starts at the bottom of column 2 and continues at the top of column 3 as “Navigation and Editing Cont’d”. All chords stay on page 1, which the compilation assert checks. The reference remains three A4 pages with every continued heading starting its page column.

Printable quick-reference digits grade 1 rename

The first digit table’s heading changes from “Digits · UEB 1” to “Digits · UEB Grade 1”. The data module and the 8 dot digit table are unchanged.

Printable quick-reference sequence space parameter

The cell renderer factors the bordered key box out of the space-key helper into a general key-box function, and defines the space key as key-box[␣]. The sequence renderer gains a space parameter: auto (the default) keeps the bordered ␣ key box between words, while any other content — such as a fixed h(6pt) gap — replaces it. Tables keep the default.

Printable quick-reference Quick Braille Fox subheads

Each Quick Braille Fox heading gains a centred three-column grid subhead: the plain text “qk brl fox”, a middle dot, and the sequence ⠿⠇ ⠍⠊⠙ ⠍⠕⠭ rendered by the sequence cell renderer with space: h(6pt) so word gaps are plain horizontal space instead of key boxes. Rendering the cells through the sequence SVG renderer — rather than as Unicode braille text — also fixes the cells showing question-mark glyphs. A linebreak before each heading keeps it off the previous content’s line, and the headings are coloured: teal on page 1 and blue on page 2, matching the page section headings.

Printable quick-reference capitals key-box intro

The Capitals intro sets “8 Dot” inside the white bordered key box and capitalises the instruction: “Add dot 7 to each uppercase letter.”

Printable quick-reference Quick Braille Fox code label

The Quick Braille Fox subheads drop the middle-dot separator and set “qk brl fox” as inline raw (monospace) code. The subhead grid becomes two columns: the code label and the braille cell sequence.

Printable quick-reference Quick Braille Fox key-box label

The Quick Braille Fox subheads set the monospace “qk brl fox” label inside the same white bordered key box used for “8 Dot” and the space key.

Printable quick-reference scalable key box

The key-box helper gains a size parameter (default 7pt) that scales the text and the box padding together. The Quick Braille Fox subhead labels use 10pt so the boxed “qk brl fox” matches the height of the braille cells beside it; “8 Dot” and the space key keep the default 7pt.

Printable quick-reference Quick Braille Fox label typeface

The Quick Braille Fox subhead labels drop the monospace raw styling; “qk brl fox” now sets in the regular bold sans face inside the key box, matching the “8 Dot” and space-key labels.

Printable quick-reference Quick Braille Fox label size

The Quick Braille Fox subhead labels return to the default 7pt key-box size; at 10pt the sans label dwarfed the 9.2pt cells. The key-box size parameter remains available for future callers.

Printable quick-reference Quick Braille Fox bare cells

The Quick Braille Fox subheads drop the “qk brl fox” label, its key box, and the two-column grid; each subhead is now just the centred braille sequence. Word gaps widen from the fixed 6pt to three non-breaking spaces ([ ]) plus the usual cell gutter.

Published Computer8 mapping

Task 9.3.2.1.1 publishes all 256 dot patterns in `references/research/computer8.yaml`: 94 printable non-space ASCII entries, dot 7 Delete and dot 8 Return command positions, and 160 reserved entries including zero. The 16-by-16 matrix is indexed by mask high and low nibble. Braille-space-only (0x100) emits literal ASCII space; zero dots is not a chord.

The verified source is VoiceOver's installed Liblouis U.S. English (EBAE) eight-dot table on macOS 26.6.2, registered as `en_US-comp8-ext.tbl`. All printable ASCII definitions match the original `en-us-comp8.ctb` fixtures. File fingerprints, Liblouis 3.36.0 provenance, and version-pinned upstream links are recorded in the research. This is not the legacy Duxbury US variant: that provider's underscore uses dots 4,5,6,7, whereas this selected Liblouis table uses 4,5,6.

Dots 7 and 8 are independent bits, not prior UEB indicators. Uppercase letters add dot 7 in the same cell; no printable ASCII character uses dot 8. Backtick is dot 4 and vertical bar is dots 1,2,5,6. Nonzero reserved cells, including dots 7,8 together, send nothing and use the dropped-prefix notice hook; phase 05 supplies its brief visual presentation. User overrides can bind them. Dot 7 and dot 8 alone belong only to the command table.

The published matrix, 94 ASCII fixtures, all 159 enterable reserved fixtures, and installed VoiceOver definitions pass the mapping checks. This section publishes the mapping, not its firmware implementation or physical acceptance.

Implemented Computer8 character table

Task 9.3.2.1.2 implements the 94 printable non-space ASCII cells as common single-cell, single-action rules. Uppercase and shifted punctuation use ZMK's implicit Shift; every row has any predicates/word position and the neutral class. Space and dot-7/Delete and dot-8/Return remain commands. Reserved entries have no factory rule, so the shared matcher emits nothing and calls the existing notice hook. No UEB or eight-dot special case is added to capture, matching, or dispatch.

All 256 patterns and literal Braille space pass end-to-end host checks, including exact reserved notices, recovery, every enterable reserved cell bound by an override, and a stream of all 94 characters with no factory buffering. Original YAML character, reserved, and isolation cases pass in split/non-split and optional Grade-2 builds, including pinned ZMK keycodes.

Native ZMK command acceptance now uses real Computer8 letters instead of fixture overrides and adds real HID case, digit, backtick, bar, underscore, shifted-symbol and reserved-cell checks. Both command tables, sticky modifiers/Globe and timeout checks pass. QBF left, Toucan left and the Toucan Computer8 boot-mode smoke builds pass. This supersedes the previous mapping-only implementation note; display presentation and physical acceptance remain unclaimed.

Computer8 isolation, switching and coverage

Tasks 9.3.2.1.3 and 9.3.2.2 verify every real Computer8 character in all 12 seeded numeric/caps/word-start combinations: characters stay immediate and neutral. Context seeding is test-only; phase-04 UEB transitions remain unimplemented. Both directions of real-mode selection discard pending prefixes, reset context and retain the shared override data. Failed name selection changes neither input nor context.

Mapped/reserved single-cell overrides and multi-cell overrides work in both modes; only explicit longer overrides add prefix delay. All real commands and Braille-space/Tab/Return whitespace paths remain available. The shared `references/key-coverage.md` records all 95 printable US ASCII characters: Computer8 covers all 95, while the current UEB1 subset covers 55 and leaves 40 contextual/fallback characters open for later phases.

QBF gains a test-only Computer8 boot-mode smoke target, matching Toucan's existing override. Both layouts parse correctly in Computer8; both real character tables link in all four central-half builds. QBF and Toucan UEB1 and Computer8 UF2 builds, native ZMK HID/command acceptance and all 41 host tests outside the two pre-existing failing cheatsheet/coordinator suites pass. Production keyboard defaults/pins are unchanged; no device was flashed. Phase-03 software is delivered, but phase 03 itself remains started pending the user-owned hardware acceptance. Visual feedback, settings and persistence remain later work.

Implemented contextual UEB Grade 1

Task 9.4 completes `ueb1` as contextual uncontracted UEB on the one shared matcher, action path and capture path. The researched indicator-context representation is restored as `references/research/ueb-indicator-context.yaml` with the numeric, capitalisation and Grade 1 research files; the class transition table in it is now engine behaviour, not future design.

Contextual rule group	Input	Result
Numeric indicator	dots 3456 then a–j	digits 1–0; sets numeric mode
Decimal comma / point	dots 3456 then dot 2 / dots 256	, / . inside numbers
Digit run	a–j while numeric mode is on	digits continue without a new indicator
Numeric space	dot 5 then a–j in numeric mode	space, then the digit; mode stays on
Capital letter	dot 6 then a–z	uppercase letter; ends word capitals
Capitalised word	dot 6, dot 6	letters capitalised until space, punctuation, a single capital, or the terminator
Capitalised passage	dot 6, dot 6, dot 6	capitalisation across spaces, punctuation and numbers until the terminator
Capitals terminator	dot 6 then dot 3	ends word and passage capitals; silent
Grade 1 indicators	dots 56 once, twice, three times; dots 56 then dot 3	silent; end numeric mode and word capitals; the next sign does not begin a word
Opening double quote / question	dots 236	" at a word start; ? elsewhere

Closing double quote	dots 356	"
Two-cell double quotes	dots 45 then 236 / 356	"
Single quotes	dot 6 then 236 / 356	'
Nondirectional double quote	dot 6 then 2356	"
Vertical line	dots 456 then 1256	|
Grave accent alone	dots 46 then 16	`

Authority: ICEB 2024 sections 2.6, 3, 4.1, 5.2–5.11, 6, 7.5–7.6, 8.3–8.7, 10 and 15.1 and Appendix 3, printed pages as recorded per sign in the research files. Uppercase and shifted characters use ZMK implicit Shift, so output stays ordinary host keystrokes. Quotes are straight US keyboard characters, not typographic Unicode.

Word context follows rule 2.6.2: space, hyphen, and the opening brackets, quotation signs and apostrophe preserve the start-of-word state that makes dots 236 read as an opening quote; letters, digits and other punctuation clear it. The grave accent modifier (dots 45 then 16) stays deliberately unmapped so it cannot be confused with the standalone backtick sign.

Implemented context transitions and pending lifecycle

The engine now applies a per-class transition after every commit: letters clear numeric mode; digits set it; numeric punctuation keeps it; space and hyphen end word capitals and set word start; openings keep word start; caps indicators set word, passage or terminator scope; Grade 1 indicators end numeric mode and word capitals; neutral rows (Computer8 and user overrides) leave context untouched. Whitespace commands share the space transition, so Return and Tab preserve a capitalised passage.

Word-position constraints are enforced with lookahead rather than timers: word-start rules need the word-start context, word-end rules wait for a following whitespace or hyphen chord, and word-middle rules need a following non-boundary character. Lookahead cells are never consumed or buffered past the eight-cell bound. A complete fallback commits the longest valid prefix and reprocesses the remaining cells exactly once in the newly committed context; command chords discard prior pending cells without host output.

Pending cells and their update, commit and discard events are exposed to the phase-05 display. Resets from layer changes, non-Braille keys, mode changes, settings entry, whitespace-preserving commands and partial dispatch failures are silent and cannot resurrect stale input. A read-only ZMK event listener resets context on layer transitions and non-Braille presses while honouring transparent layers, custom Braille devices, generated host keystrokes and press-time release ownership.

Contextual UEB1 verification boundary

All 456 existing UEB1 acceptance fixtures, all 222 recovered rulebook and labelled constructed examples, and an exhaustive 95-character US keyboard audit pass through the real C capture, registry, matcher and dispatcher, in split and non-split builds with and without optional Grade 2 compilation and against the pinned ZMK keycode header. Native unmodified ZMK acceptance now also covers contextual HID output, layer and non-Braille event boundaries, and the transparent and custom keymap nodes; routing and mode/settings native suites pass

with engine symbols wrapped instead of redefined. QBF and Toucan UEB1 and Computer8 central-half UF2 builds pass against clean HEAD snapshots of both keyboard repositories. `references/key-coverage.md` now records both modes covering all 95 printable characters in software. No physical keyboard or VoiceOver acceptance, display feedback, settings UI, persistence, or Grade 2 support is claimed.